

## SMART STUDENT HUB SYSTEM

### ROLE: SOFTWARE PROGRAMMER

#### Overview

Software programmer is a professional who writes and create codes or instructions that tells the computer what to do, using programming languages such as Java, Java Script, C++ and Python. The programmer plays vital role in Smart Student Hub software development by transforming ideas into code, test and maintain code to develop functional software systems project.

#### Key Responsibilities

1. **Software development** - the software programmer write code using programming languages to develop and bring the software system project into action. Writing a clear and efficient code using HTML, CSS and Java Script developing core features that include: Login System, Dashboard Interface, Task and deadline tracker
2. **Design Implementation** – convert the design system into user-friendly interface, ensuring that the visual appearance of the system is appealing and easy to use by applying responsive design principles for accessibility.
3. **Functionality Implementation** – ensuring that all system features work correctly and meet the project objectives such as: adding, updating and deleting tasks, displaying deadlines,
4. **Debugging and testing** – identify and fix errors in the Smart Student Hub system, as if there are errors in the system it may malfunction and produces incorrect output. Then perform testing to ensure that all features function properly and to improve system performance and reliability.
5. **Integration** - work closely with the System Architect to integrate different components
6. **Version Control Management** – uploading and managing code on GitHub and collaborate with team members using branches and commits
7. **Documentation** – to explain how the code works, to make it easier for others to understand the code so that they can maintain the system, explain to others who have interest in coding, report if there are any changes observed in the system
8. **Collaboration** - work with team members to achieve project goals, participating in group meetings and provide progress updates and assist others with technical issues when necessary

**Expected Outcome:** A fully functional Smart Student Hub System. Clean, well-structured and maintainable code. Successful implementation of all required features

#### Reference

Acuna, S.T. & Juristo, N. (2004), Assigning people to roles in software p and projects. Software: Practices and experience, 34(7):675-696.